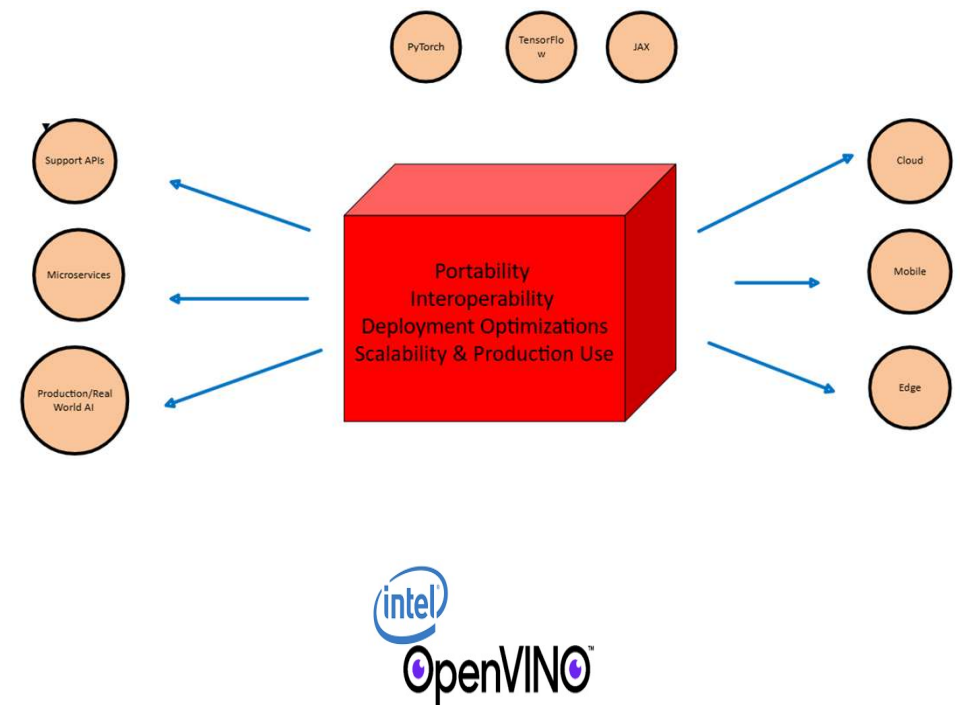


Model Exporters and Quantization Workflows

Sahar F., Yamini N, Garth, Ryan, Anirudh, Freddy

Exporters

- ONNX – Cross Framework Compatibility
- OpenVINO – Intel Hardware Optimization
- CoreML - Apple Ecosystem
- TFLite/TensorFlowSaved Model Mobile Edge Devices
- Torch Script/Torch Dynamo PyTorch Production
- ExecuTorch Tight Integration with Pytorch Environment. Mobile / Embedded Systems.



Tool Chains Optimum/Olive/ORT Gen AI

Optimum Support Good For Developers and Prototypes.

- Exporters to ONNX and OV IR.
- Optimization with ONNXRT, OnnxScript
- Quantization with NNCF-OVIR, ORT Quantizer-ONNX.
- Inference Pipeline.

ORT GenAI Model Builder Direct Integrated with ORT Gen AI. Good for Deployment in Production Systems

- Allows Implicit KV Cache Management Strategies.
- Quantization Support ORT Quantizer
- Optimized kernels for Ops with specific hardware, contrib ops which may not be compliant with OpenVINO.
- Optimized for Gen AI Pipeline
- Config file.json is needed to run with Gen AI Pipeline.

Olive Tool Chain

- Olive executes a *workflow*, which is an ordered sequence of individual model optimization tasks called *passes*
- Example passes include model compression, graph capture, quantization, and graph optimization example Model Builder Pass, Pytorch export, NNCF Quantization Pass
- Each pass has a set of parameters that can be tuned to achieve the best metrics, such as accuracy and latency, that are evaluated by the respective *evaluator*.

Slide 3

- SF1** [@Swaminathan, Anirudh] does olive uses optimum for export or uses its own custom scripts.
Fatima, Sahar, 2026-04-03T02:32:22.250
- 1 0** Olive for openvino exclusively uses optimum-intel expoort openvino path
Olive for ONNX has a couple of options
It has an equivalent of optimum-cli export onnx in its OptimumConversion pass
It also has ModelBuilder which utilizes ORT GenAI Model Builder capabilities
It also has PyTorch export, which has a few patches for export as well
Swaminathan, Anirudh, 2026-04-06T04:16:52.253
- SF2** [@Metcalf, Ryan] Can you please provide some info on Model Builder Specialized KV Cache Management and
ONNX Operator Specialization for GenAI Models
Fatima, Sahar, 2026-04-03T02:46:50.375

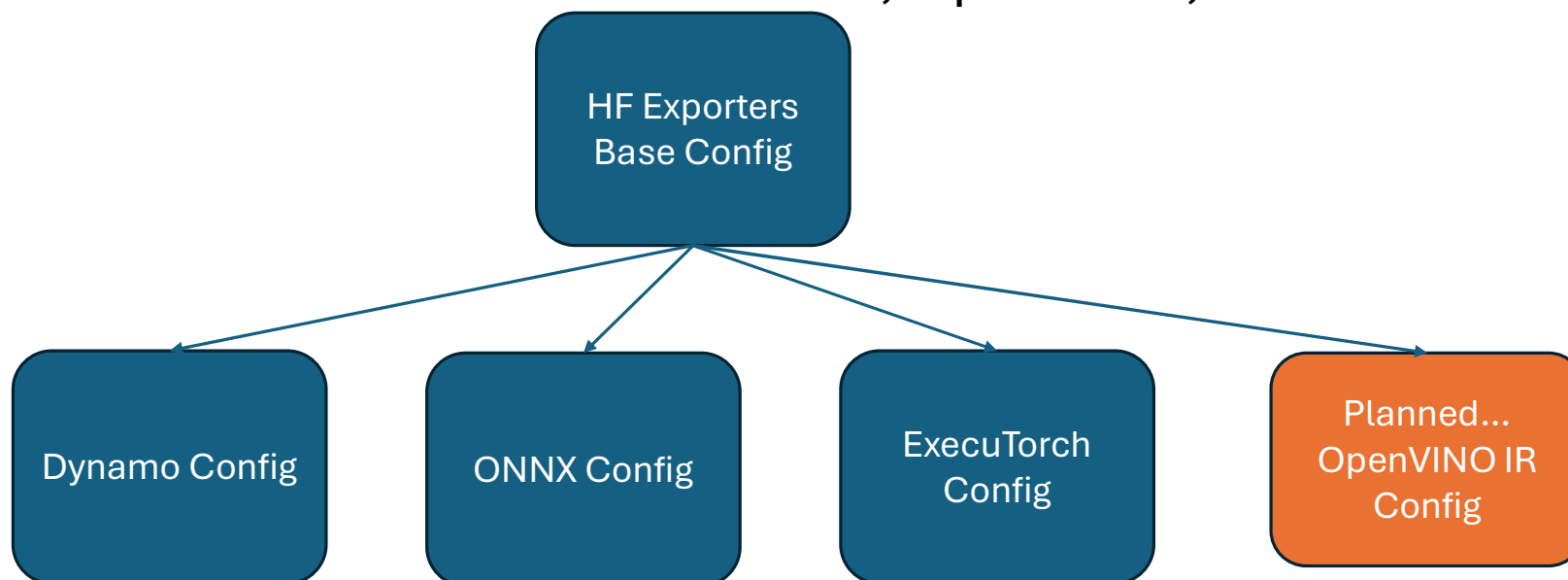
ONNX Export

- Legacy Torch-ONNX Export uses Olive , Optimum, Transformers(Hugging Face)
- Optimum is optimized for OpenVINO/Intel through direct export to Optimum-OpenVINO IR.
- Legacy export used TorchScript : Used two main methods Tracing, Scripting . `Torch.onnx.export()` call used torchscript
- PyTorch is phasing out Torch Script in favor of more robust JIT Compiler TorchDynamo which intercepts PythonBytecode into a FX Graph, employs guard mechanisms and handle salient features like Symbolic/Dynamic Shapes [`torch.export.export\(\)`](#)

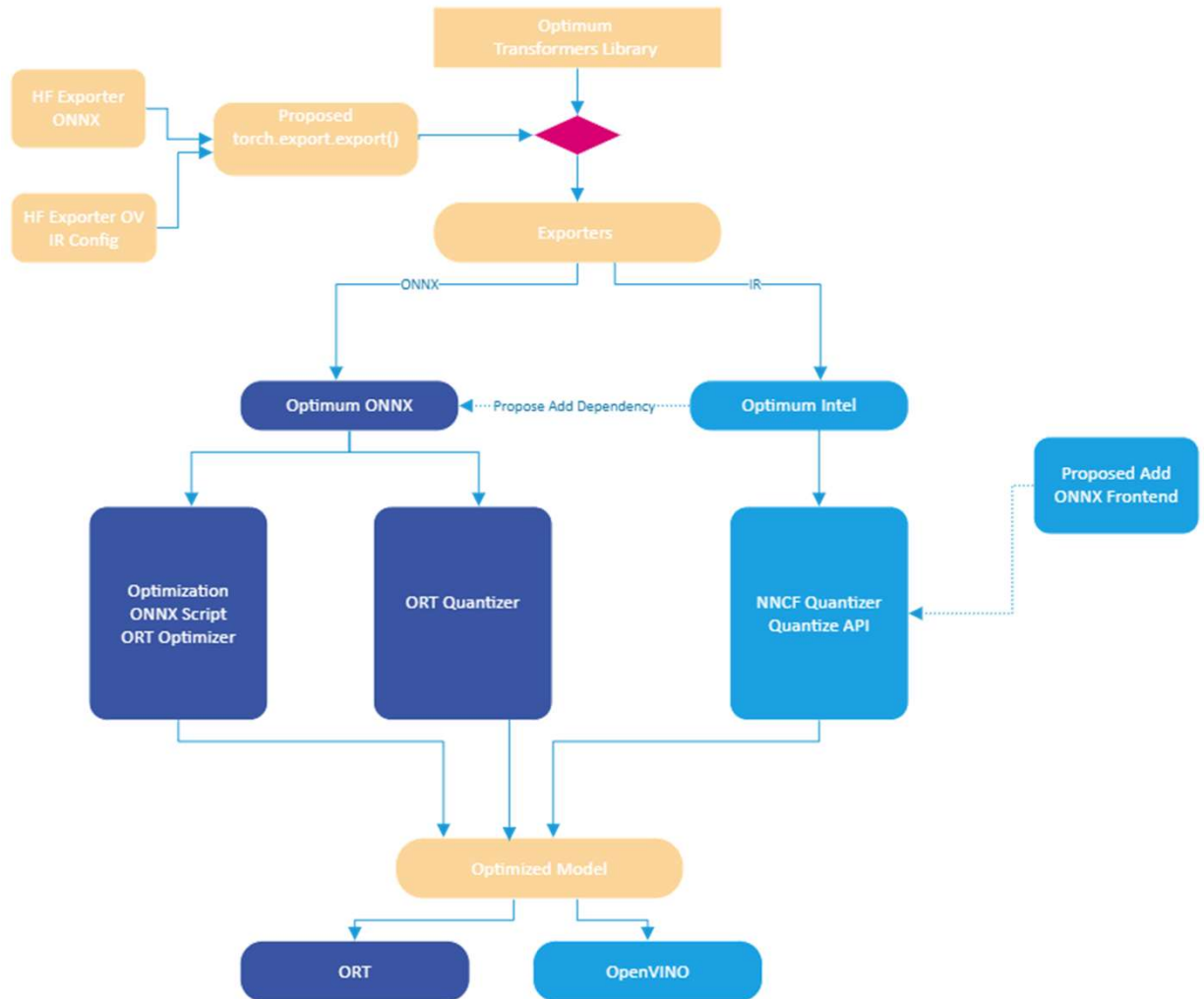


Transformers HF Exporters () In Progress

- HF Exporters() is a new feature introduced which uses Torch Dynamo Export to centralize the export functionalities of Transformer Libraries for ONNX, OpenVINO, ExecuTorch



Optimum Export Workflow



Quantization Workflows and Tools

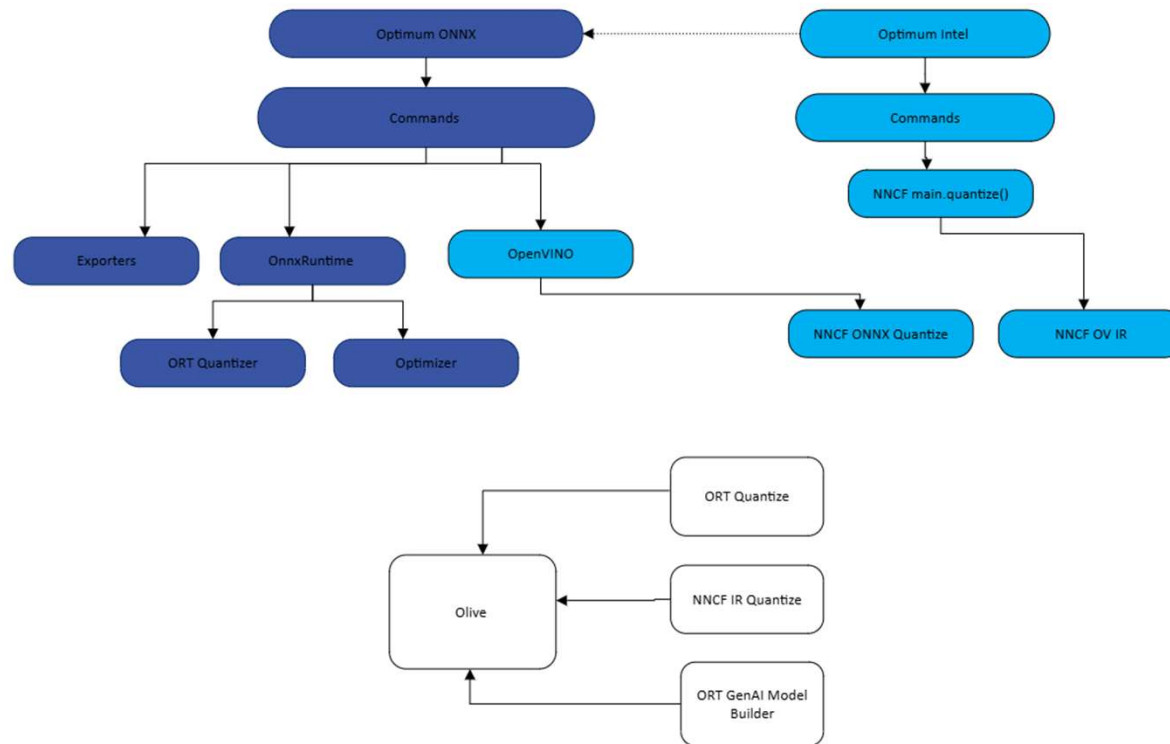
- Gen AI Pipelines, Export Pipelines(Optimum/Olive) integrate Quantization and Optimization Tools in the export pipelines
- Intel Optimized Quantization Tool NNCF Supports Intel Optimized Hardware, Multiple Compression techniques, Frontends(ONNX, Pytorch) and Backend(ONNX, Open VINO IR)

ORT Quantizer
Dynamic
Quantizer
Static Quantizer
Compression
Aware

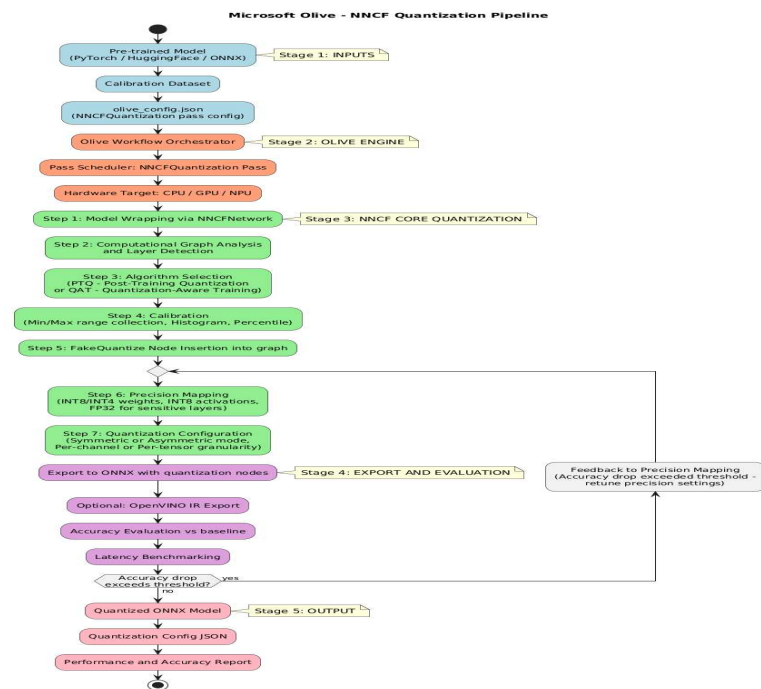
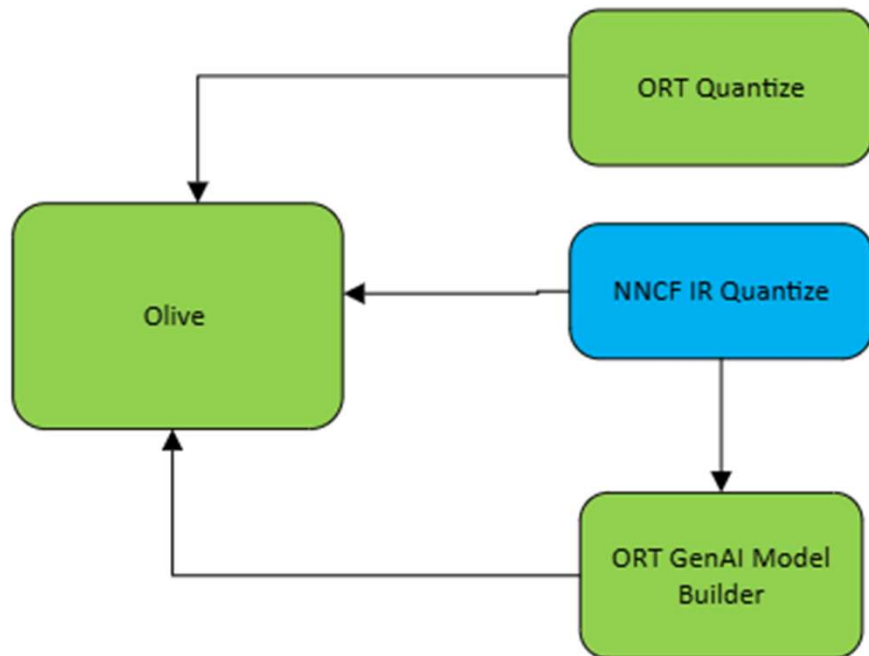
NNCF
PTQ
QTA
Weight
Compression,
Pruning...

TensorFlow/Pyt
orch PTQ and
QTA
Quantization

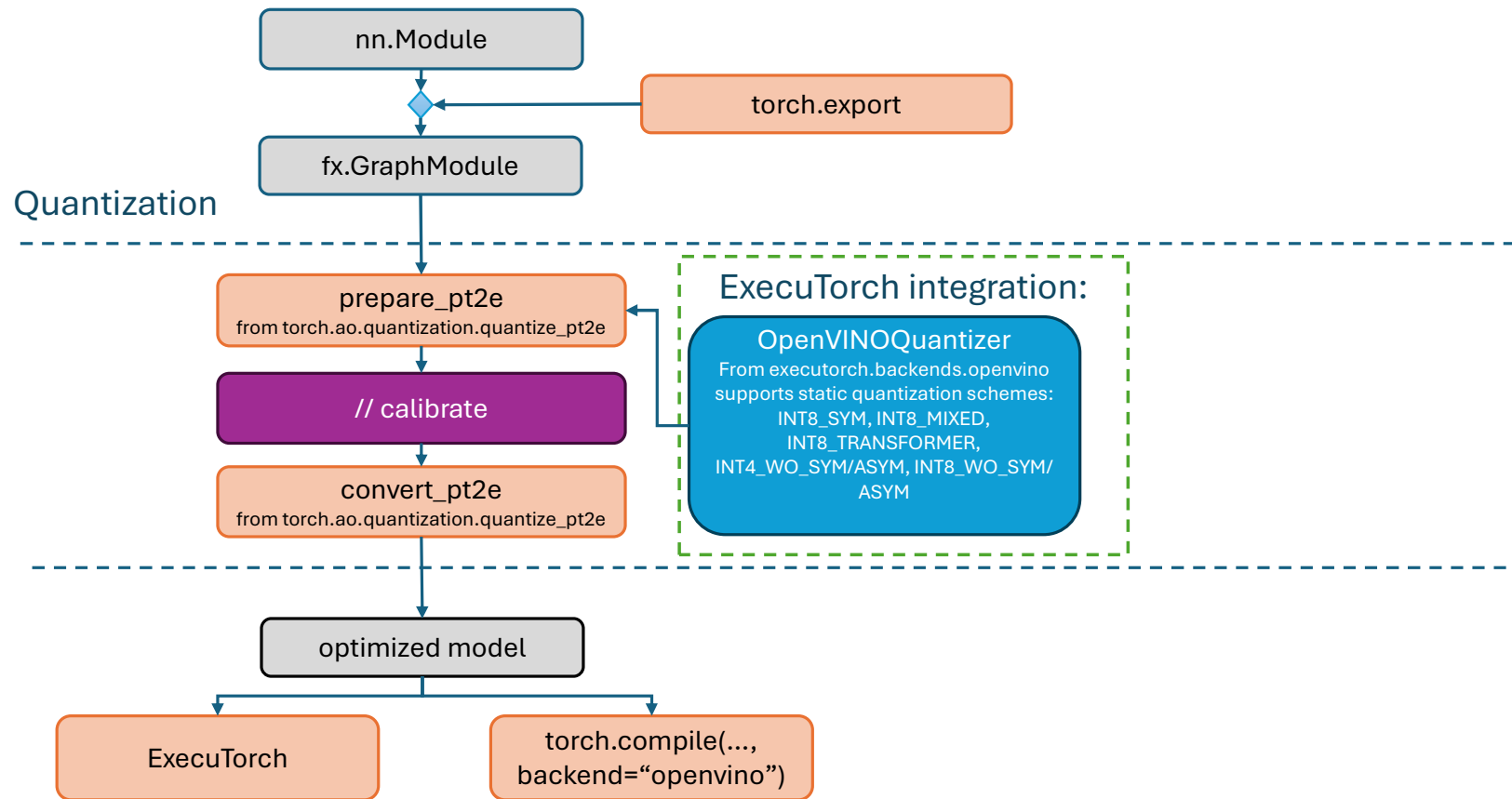
Quantization Workflows – Optimum/Olive



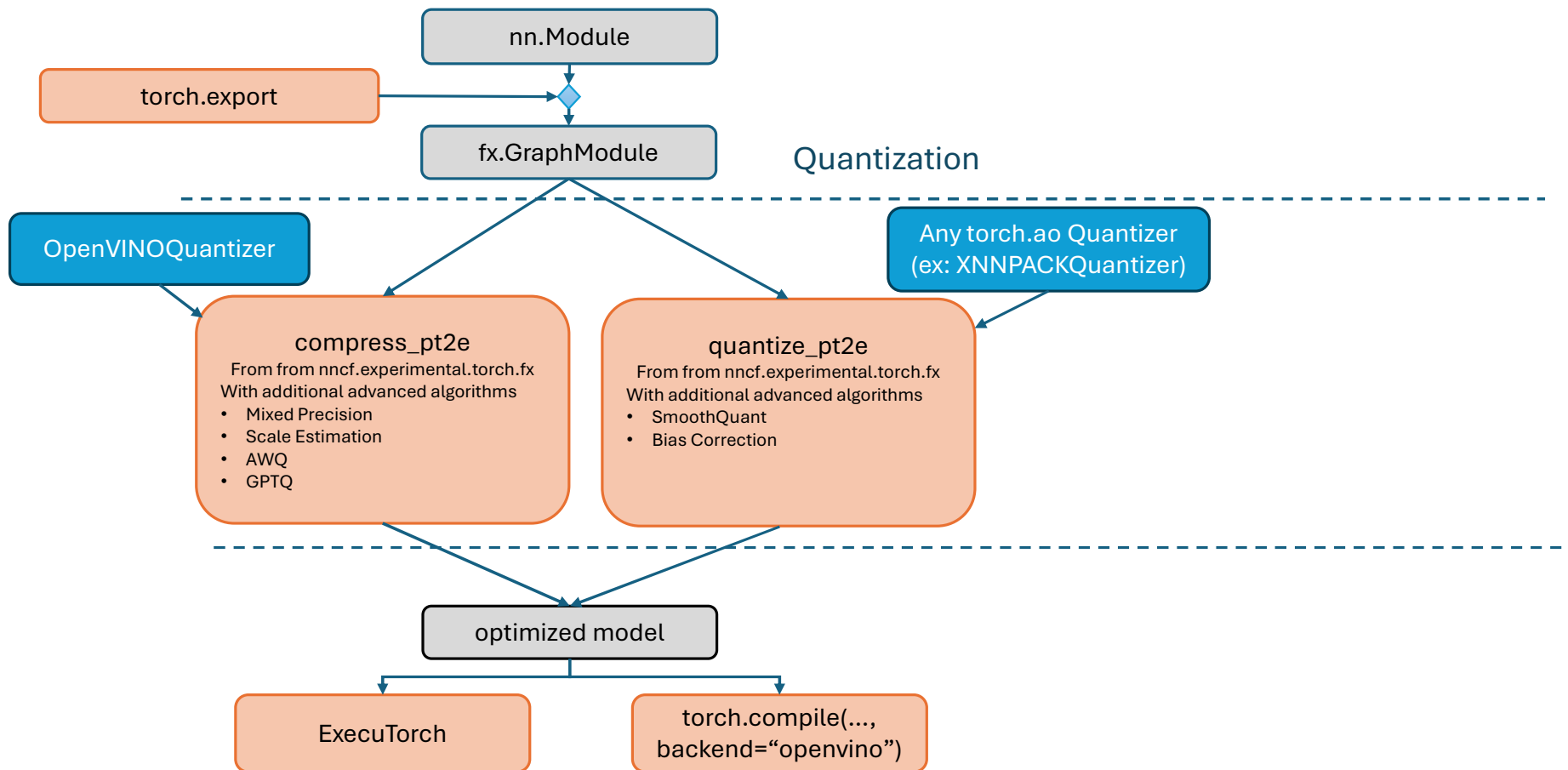
NNCF-Olive Integration



Executorch OpenVINOQuantizer

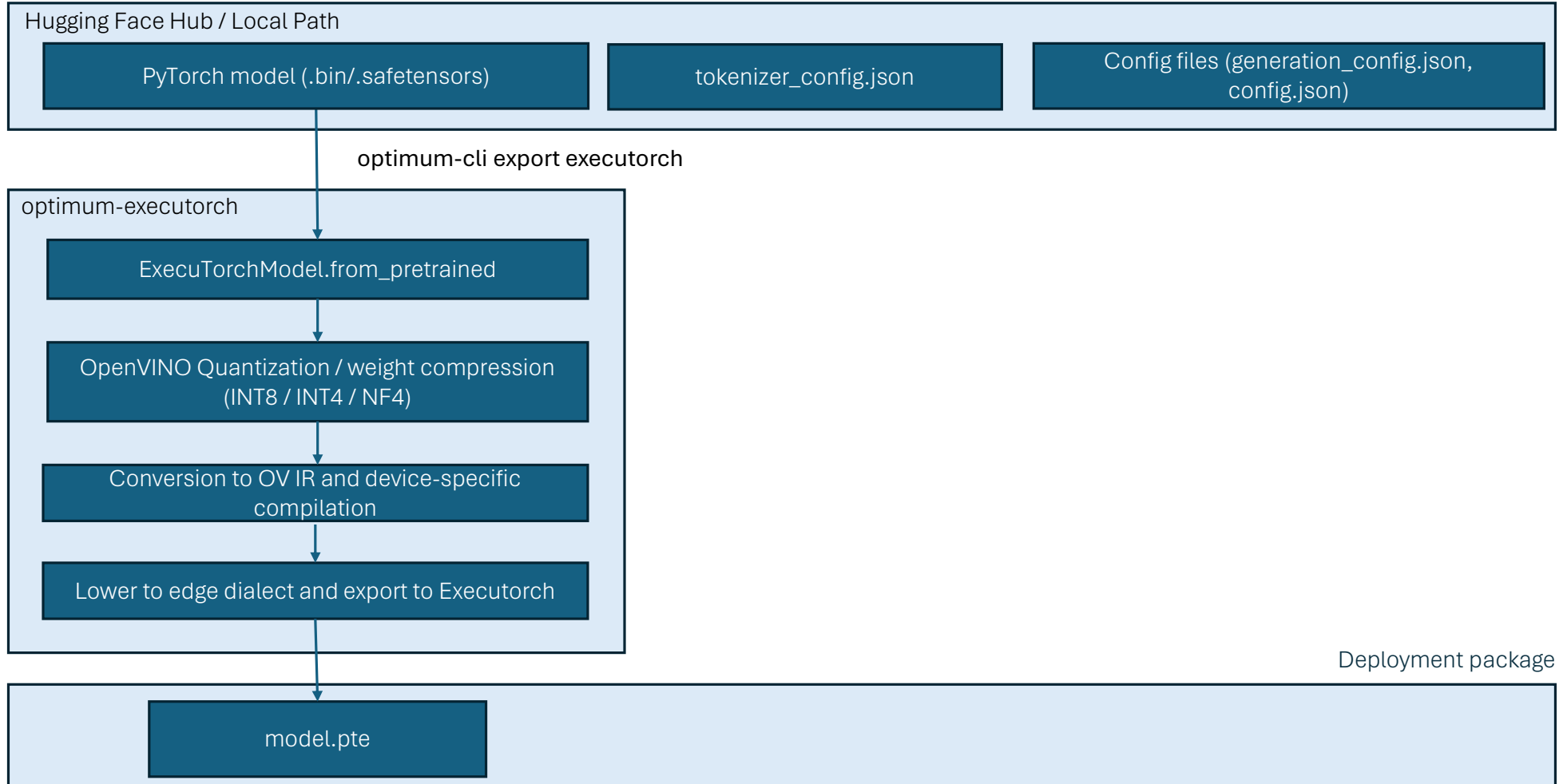


NNCF Executorch Compression API



Optimum-ExecuTorch Integration

Initial support : https://github.com/suryasidd/optimum-executorch/tree/openvino_recipe



Proposed Solutions - Status

- Integrate NNCF with AI Framework Quantization Workflows. NNCFIR – Olive. NNCF ONNX Integration with Optimum pending
- Reuse Export mechanism across AI Frameworks. Hugging Face Exporters()
- Upload Intel Optimized Models in Model Zoos and HF Repo for easy ISV Use.
- Upload Samples for Export and Quantization Workflows. Innersource repos with custom scripts. External Facing Samples
- Open Issues:
 - Dynamo Maturity
 - GenAI Config File needed for ORT GenAI.
 - Contrib Operator Usage.

Model Zoo and Samples Repos

- OpenVINO <https://huggingface.co/OpenVINO/collections>
- [ONNX Model Zoo](#).
- [Notebooks](#)
- [Dynamo Exported Custom Scripts](#)

Slide 14

- SF1** [@Nimmagadda, Yamini] [@Chiu, Freddy] Please add any Repos/examples for Samples of ONNX, Pytorch Export and Quantization Workflows that you may be aware of
Fatima, Sahar, 2026-04-06T07:14:54.376
- FC1 0** ORT model builder
Chiu, Freddy, 2026-04-08T04:42:53.301
- SF2** [@Swaminathan, Anirudh] Are there any Olive Integration Workflows samples that can be tagged here.
Fatima, Sahar, 2026-04-06T07:16:55.539
- 2 0** Yes, there are a lot of OpenVINO model recipes in the official MSFT Olive-recipes repo
<https://github.com/microsoft/olive-recipes>
Swaminathan, Anirudh, 2026-04-06T15:59:16.443